

Math 133 Written Homework 2

Due Wednesday, September 16, 2026, at 11:59 p.m.

Submit one legible PDF on Gradescope, clearly identify each problem, and assign the correct page(s) to every problem. Show enough reasoning that another student in the course could follow your work, and include units whenever they are meaningful. You may discuss problems with classmates, but your submitted work must be your own. Late work will not be accepted.

Except for the Excel problem, draw graphs by hand, label axes and important points, and show the calculations used to make each graph. You may use technology to check your work. Use algebra and graphs; derivatives are not needed. Give intervals of increase and decrease in interval notation, using open intervals.

0. Assignment record.

- (a) List any collaborators and the problem(s) you discussed with each person. If none, write “None.”
- (b) How much time did you spend on this assignment?
- (c) Disclose any use of generative AI: name the tool and describe how you used it. If none, write “None.”

1. **A shipping fee that jumps.** A shipping company charges $S(w)$ dollars to ship a package weighing w pounds:

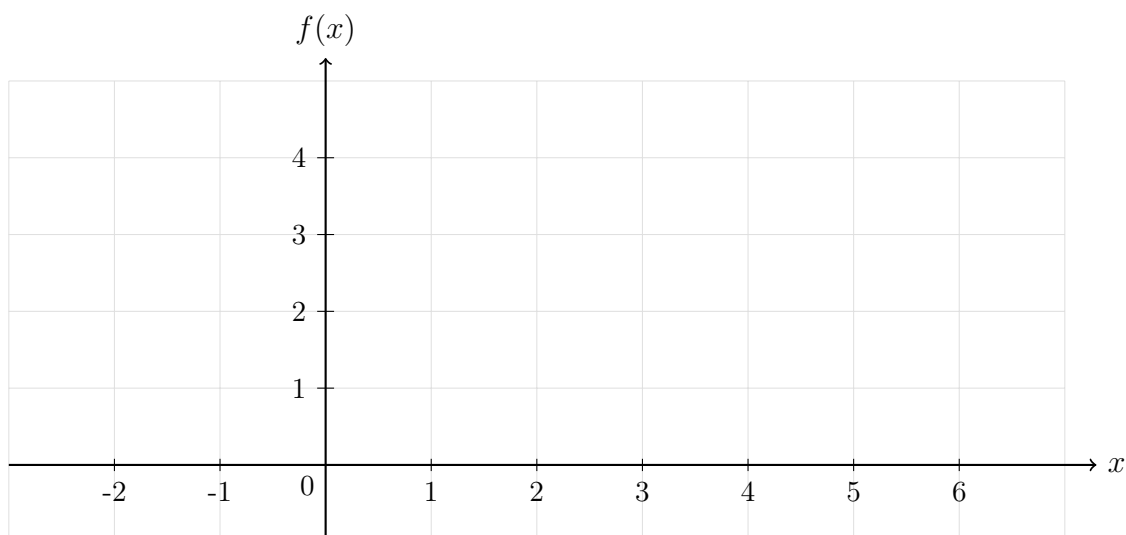
$$S(w) = \begin{cases} 4 + w, & 0 < w \leq 4, \\ 9 + \frac{3}{4}w, & 4 < w \leq 8. \end{cases}$$

- (a) Find $S(4)$ and $S(5)$. Sketch the graph, using open and filled dots to show what happens at $w = 4$.
- (b) Compute $S(5) - S(4)$, $S(2) - S(1)$ and $S(6) - S(5)$. Describe in words what each of these quantities means in the context of shipping a package.

2. **Graphing one piece at a time.** Let

$$f(x) = \begin{cases} x + 2, & -2 \leq x \leq 1, \\ 3, & 1 < x \leq 3, \\ 6 - x, & 3 < x \leq 6. \end{cases}$$

- (a) Calculate $f(-2)$, $f(0)$, $f(1)$, $f(2)$, $f(3)$, and $f(6)$. Beside each calculation, identify which formula applies.
- (b) Graph f on the grid. Draw each rule only over its stated interval. Use filled dots for included endpoints and open dots for excluded endpoints; where pieces meet at an included point, show a filled dot.
- (c) State the intervals where f is increasing, decreasing, and constant. Give its range and its maximum value, including every x at which the maximum occurs.



3. **Quadratic graphs and restricted intervals.** Let $h(x) = 9 - (x - 2)^2$ and $k(x) = x^2 - 6x + 5$.

- (a) Sketch both functions on separate axes. Label each vertex and all intercepts. You may use $x = -b/(2a)$ to locate the vertex of $ax^2 + bx + c$; plot additional points as needed.
- (b) For each function, state the intervals where it is increasing and decreasing on $\mathbb{R}$.
- (c) For *each* function, find its maximum and minimum values on $[0, 4]$, and give every x where each occurs. Mark the endpoints and compare their heights with the vertex.
- (d) Repeat part (c) on $[4, 6]$. Explain how excluding the vertex changes where the maximum or minimum occurs.

4. **A nonlinear demand model.** A supplier models the price per kilogram of a product by

$$p(q) = \frac{120}{q+2}, \quad 0 \leq q \leq 10,$$

where q is the number of kilograms sold and $p(q)$ is in dollars per kilogram.

- (a) Calculate $p(0)$, $p(2)$, $p(4)$, and $p(10)$. Is the price increasing or decreasing on this domain? Explain directly from the formula.
- (b) Write the revenue function $R(q) = qp(q)$ and calculate $R(0)$, $R(2)$, $R(4)$, and $R(10)$. Does a decreasing price necessarily mean decreasing revenue? Use your values to explain.

5. **An exponential depreciation model.** A bakery buys a delivery van for \$24,000. Its resale value is modeled by

$$V(t) = 24000(0.80)^t, \quad 0 \leq t \leq 6,$$

where t is time in years after purchase and $V(t)$ is measured in dollars.

- (a) Interpret 24,000 and 0.80 in this model. What percentage of its value does the van lose each year?
- (b) Calculate $V(0)$, $V(1)$, $V(2)$, $V(3)$, and $V(6)$, rounding to the nearest dollar. Sketch the model over $[0, 6]$ and identify its maximum and minimum values and when they occur.
- (c) Compare the dollar loss during the first year with the dollar loss during the second year. Does the model predict the same dollar loss each year? Explain why this is an exponential model rather than a linear one.

6. **Excel investigation: choosing an advertising budget.** A small business models the profit from a promotion by

$$P(x) = 1800(1 - e^{-0.4x}) - 100x - 15x^2 - 200, \quad 0 \leq x \leq 10.$$

Here x is the advertising budget in *hundreds of dollars*, and $P(x)$ is profit in dollars. The first term models revenue from the promotion; the remaining terms account for advertising, other operating costs, and a fixed setup cost. For example, $x = 3$ means an advertising budget of \$300. Use Excel to estimate the advertising budget that maximizes profit *graphically*. The notation e^u is entered in Excel as EXP(u).

- (a) **Make a table.** In A1 enter Budget (hundreds of dollars) and in B1 enter Profit (dollars). Fill A2:A12 with 0, 1, ..., 10. In B2 enter

$$=1800*(1-EXP(-0.4*A2))-100*A2-15*A2^2-200$$

and fill down through B12. Format profit to two decimal places.

- (b) **Graph the whole interval.** Create an XY scatterplot with connected markers, with x on the horizontal axis and $P(x)$ on the vertical axis. Include a title, units, and the full interval $0 \leq x \leq 10$. Estimate the location and value of the maximum from the graph. Explain why neither endpoint gives the maximum.
- (c) **Look more closely.** Make a second table using increments of 0.1, starting one unit below your best whole-number x and ending one unit above it. Graph these values in a second XY scatterplot with connected markers. Use this graph and its plotted values to refine your estimate. Report the advertising budget to the nearest \$10 and the corresponding predicted profit to the nearest cent.
- (d) **Interpret the result.** Explain why your answer is an estimate rather than an exact maximum. According to the graph, what happens to profit if the business keeps increasing its advertising budget past your chosen amount?

Include with this problem in your Gradescope PDF: screenshots of both tables and both labeled graphs, the formula entered in B2, and your written answers. Make the values near the estimated maximum readable.